Pharmacology of Immunosuppressant Drugs, Part 1

- **Overview of transplantation immunosuppression**
- **Lymphocyte depleting and nondepleting agents for transplantation induction therapy**

Leslie Goldstein, Pharm.D.
Associate Professor
Department of Clinical Specialties
Division of Pharmacology
lgolds01@nyit.edu

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

I am available to groups and individuals for pharmacology help and discussions by appointment.

lgolds01@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

After completion of the preparation materials, students should be able to:

1. List the class pharmacokinetic properties of the antisera, the monoclonal antibody therapeutics, and glucocorticoids.
2. Relate the mechanisms of action of each class of drugs to their effects on the suppression of cell-mediated and humoral immunity and give specific examples.
3. Describe the mechanisms of toxicities and drug interactions and give specific examples.
4. Prepare a concept map that bridges the chemical or biological structures of each class of immunosuppressants to its pharmacokinetic properties, mechanisms of action, therapeutic uses in transplantation pharmacology, and toxicities.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Highly relevant optional materials:

- Video Lecture | Dr. Goldstein's Notes Handout | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links

Name of Discipline/ Organ System: Immunology > Treatment of Immune Disorders
Title of Brick: Immunosuppressants
Link: https://exchange.scholarrx.com/brick/immunosuppressants

Access Medicine Goodman & Gilman’s The Pharmacological Basis of Therapeutics, 15e, 2023; Chapter 39: Immunosuppressants, Immunomodulation, and Tolerance > Relevant sections
https://nyit.idm.oclc.org/login?url= https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3191&sectionid=269169964

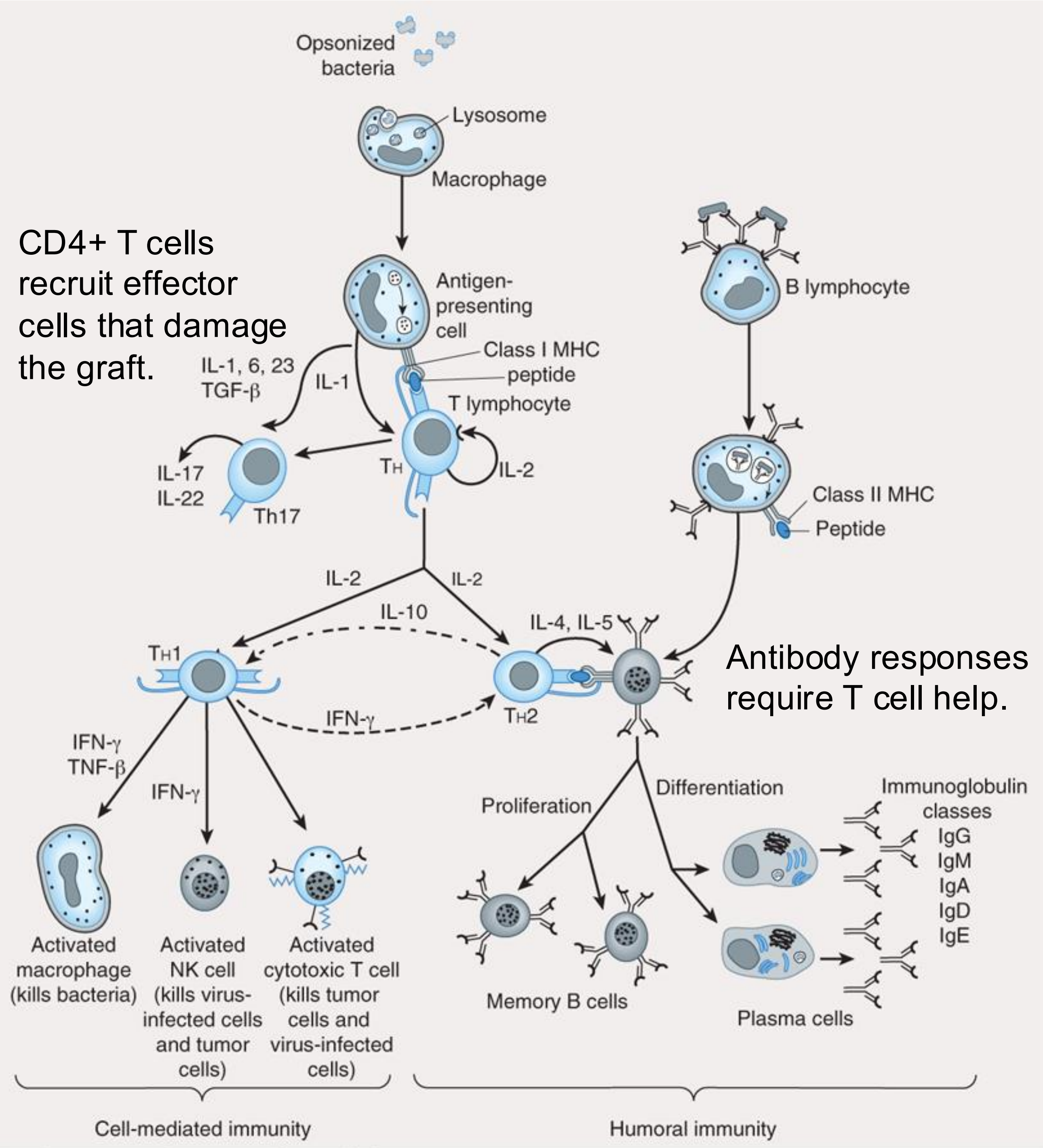
Access Medicine Katzung’s Basic & Clinical Pharmacology, 16e, 2024; Chapter 55: Immunopharmacology > Relevant sections
https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/book.aspx?bookid=3382#281742356

LWW Health Library, Premium Basic Sciences, Lippincott’s Illustrated Reviews: Pharmacology 8e, 2023; Chapter 36: Immunosuppressants
https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329619&bookid=3222

Key points

Knowledge of the biological function of T and B lymphocytes is essential for understanding the pharmacology of immunomodulation.

- This lecture focuses on the mechanisms of action and adverse effects of immunosuppressive therapies to prevent acute solid organ allograft rejection.
- The basic principles of immunosuppressants in this context can be applied to their application for autoimmune and inflammatory diseases and graft vs host disease (GVHD).
- The pharmacology of drugs that enhance the immune system is not addressed in this lecture.

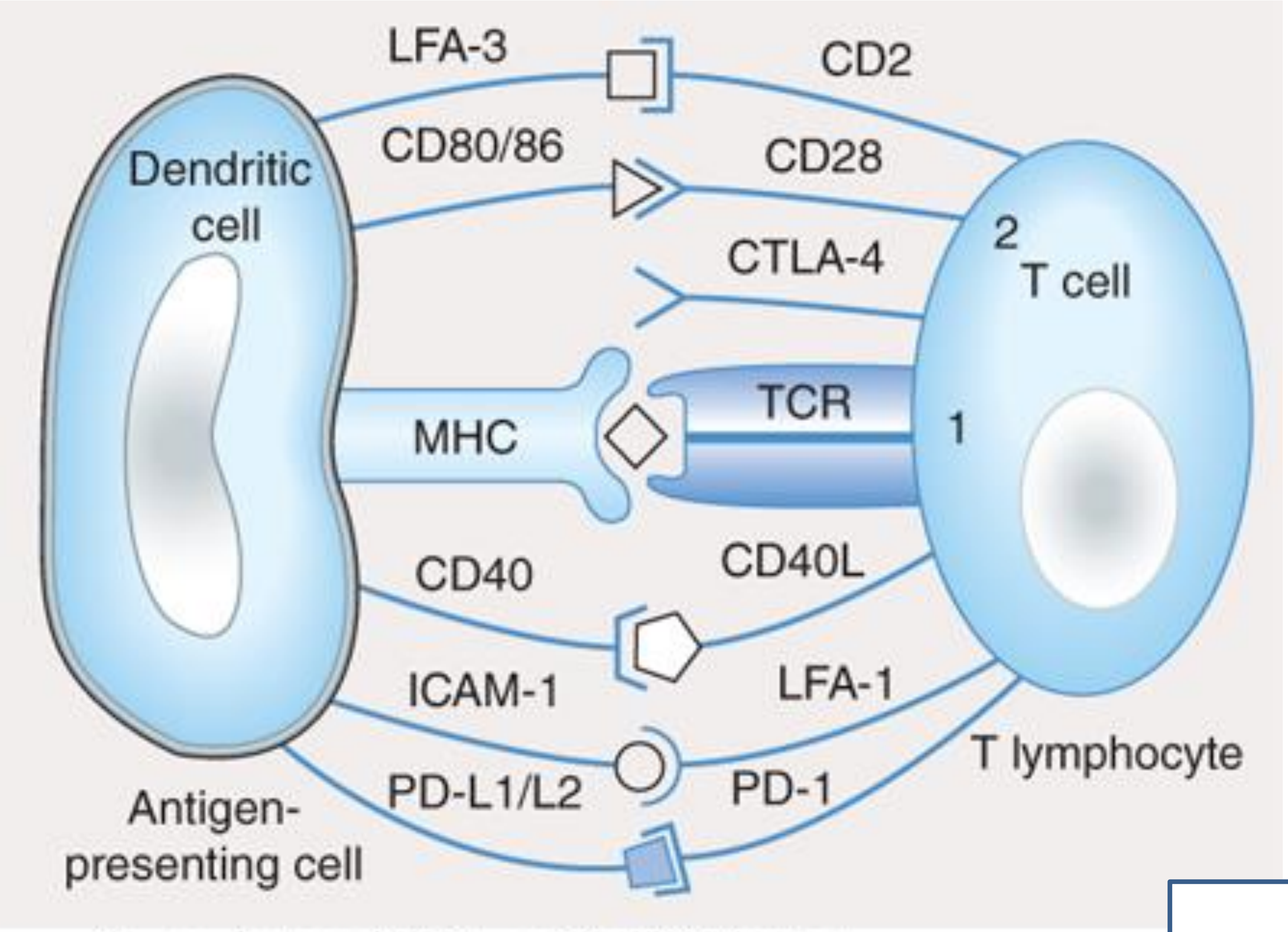


Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition
Copyright © McGraw-Hill Education. All rights reserved.

Scheme of cellular interactions during the generation of cell-mediated and humoral immune responses.

- **The cell-mediated arm** of the immune response involves the ingestion and digestion of antigen by antigen-presenting cells such as macrophages.
- Activated TH cells secrete IL-2, which causes proliferation and activation of cytotoxic T lymphocytes as well as TH1 and TH2 cell subsets.
- TH1 cells also produce IFN-γ and TNF-β, which can directly activate macrophages and NK cells.
- TH17 cells may be induced by IL-1, -6, -23, or TGF-β secretion by antigen-presenting cells; TH17 cells are inflammatory and secrete IL-17 and -22.
- **The humoral response** is triggered when B lymphocytes bind antigen via their surface immunoglobulin. They are then induced by TH2-derived IL-4 and IL-5 to proliferate and differentiate into memory cells and antibody-secreting plasma cells.
- Regulatory cytokines such as IFN-γ and IL-10 down-regulate TH2 and TH1 responses, respectively (dashed arrows).

Citation: Chapter 55 Immunopharmacology, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021 Figure 55-3. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250603682 Accessed: April 06, 2022
Copyright © 2022 McGraw-Hill Education. All rights reserved



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition
Copyright © McGraw-Hill Education. All rights reserved.

MHC peptide complex
TCR-CD3
(signal 1)
+
CD80/86 (B7)
CostimulatoryCD28
(signal 2)
↓
Required for activation

NOTE: CTLA-4 is an “immune checkpoint”. CTLA-4 interaction with CD80/86 transmits an inhibitory signal.

T-cell activation by an antigen-presenting cell requires engagement of the T-cell receptor by the MHC-peptide complex (signal 1) and binding of the costimulatory molecules (CD80, CD86) on the dendritic cell to CD28 on the T cell (signal 2). The activation signals are strengthened by CD40/CD40L and ICAM-1/LFA-1 interactions. In a normal immune response, T-cell activation is regulated by T-cell–derived CTLA-4 and PD-1. CTLA-4 binds to CD80 or CD86 with higher affinity than CD28 and sends inhibitory signals to the nucleus of the T cell, while ligation of PD-1 by PD-L1 or -L2 also inhibits T cell proliferation.

Citation: Chapter 55 Immunopharmacology, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021, 55-2. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250603682 Accessed: April 06, 2022
Copyright © 2022 McGraw-Hill Education. All rights reserved

Key points

- Immunosuppressive therapies – drugs that suppress the immune system – are important in preventing rejection of allografts and in the treatment of disorders due to immune system dysregulation, such as autoimmune and inflammatory diseases.
- Acute rejection results from expression of HLA class I and II antigens in the allograft and early infiltration of CD8+ T cells. CD4+ T cells orchestrate acute rejection by recruiting the effector cells that damage the graft, including CD8+ T cells macrophages, natural killer cells, and B cells.
- Prior to allogeneic solid organ and hematopoietic stem cell transplants, patients may receive induction therapy with an immunotherapy regimen to suppress the T cell response and prevent acute rejection. The therapy typically consists of antithymocyte globulin (rATG) or basiliximab (anti-CD25), along with high-dose methylprednisolone (a glucocorticoid). The anti-CD52 panlymphocytic inhibitor alemtuzumab is an infrequently used option for patients at high risk of acute rejection.
- Following transplantation, maintenance immunosuppression therapy is initiated to prevent acute rejection – typically the combination of a calcineurin inhibitor, an antimetabolite, and a glucocorticoid. Alternatives are the CTLA-4 inhibitor or an mTOR inhibitor.

Key points

- **Antithymocyte globulin (ATG)** consists of polyclonal antibodies prepared by immunization of horse, sheep, or rabbits with human thymocytes. Rabbit ATG (Thymoglobulin) is the most frequently used antiserum for transplant induction therapy. The antiserum contains antibodies to a wide variety of T cell antigens including MHC antigens and produces profound lymphopenia. It appears that ATG acts on T cell surface antigens, causing depletion of circulating T cells through cytotoxicity and other mechanisms.
- ATG is used as induction therapy to prevent early allograft rejection and to treat severe glucocorticoid-resistant acute rejection episodes, along with other immunosuppressants. ATG can cause infusion reactions characterized by flu-like symptoms and pain and erythema at the infusion site. Rapid infusion may lead to release of cytokines by activated monocytes and lymphocytes. Prolonged administration may lead to profound immunosuppression with an increased risk of opportunistic infections and post-transplant lymphoproliferative disorder (PTLD).
- **Basiliximab** is a monoclonal antibody directed against IL-2 receptor α chain (CD25) expressed on activated T cells, thus interferes with T cell proliferation. It does *not* cause T cell depletion. Basiliximab is administered for induction in patients at low risk of rejection, along with IV methylprednisolone and maintenance immunosuppressive combination therapy. The drug is generally well tolerated.

Key points

- **Alemtuzumab** is an anti-CD52 monoclonal antibody infrequently used in transplant induction therapy. CD52 is a cell surface glycoprotein expressed on T and B cells. It causes potent and prolonged depletion of T and B cells by complement-mediated and antibody-dependent lysis. Alemtuzumab can cause serious, even fatal, pancytopenia, infusion reactions, infections, and autoimmune reactions.
- **Glucocorticoids** are included in both induction and maintenance immunosuppressive regimens. High dose IV methylprednisolone is given in combination with rATG and basiliximab for induction and for acute rejection episodes (alone or with rATG). Oral prednisone is a component of the standard maintenance regimen – a calcineurin inhibitor, an antimetabolite, and glucocorticoid.
- Glucocorticoids bind intracellular glucocorticoid receptors, translocate to the nucleus and regulate gene expression and have a wide array of physiologic effects. Glucocorticoids rapidly reduce T cell populations by lysis or redistribution, with suppressive effects on inflammatory cytokines and chemokines. Concentrations of neutrophils, monocytes, basophils, and eosinophils are also decreased.
- The drugs for maintenance prophylaxis of acute rejection are described in Part 2.

Organ Transplantation: Allografts

- Commonly transplanted organs are kidney, liver, heart, lungs, pancreas, and intestines.
- Immunosuppressive drugs are used to dampen the immune response in organ transplantation.
- Combinations of drugs blocking different targets in the immune cascade may allow lower doses to maintain adequate immunosuppression and minimize adverse effects.
- Transplantation improves long-term survival and quality of life for many patients afflicted by organ failure.

Allograft: tissue graft from a human donor who is not genetically identical to the human recipient.

Immune reactions to foreign tissue

Allograft rejection

- Recipient’s immune system attacks foreign antigens in the allograft.

Graft-versus-host disease (GVHD)

- Immunologically competent immune cells in allograft transplanted into immunodeficient recipient recognize the recipient’s antigens as “foreign” and attack it.
- Common complication of allogeneic hematopoietic stem cell transplant.
- Can occur following transplantation of solid organs rich in lymphoid cells, such as liver, or with transfusion of unradiated blood.

Transplant rejection

Hyperacute rejection: within minutes to hours

- Preformed host antibodies against alloantigens bind grafted vascular endothelium and complement is activated → necrosis
- Occurs within hours of the transplant
- Cannot be stopped by immunosuppressant drugs

Acute rejection: days to weeks

- Cellular, humoral or both
- Acute rejection results from expression of HLA class I and II antigens in the allograft and early infiltration of CD8+ T cells. CD4+ T cells orchestrate acute rejection by recruiting the effector cells that damage the graft, including CD8+ T cells macrophages, natural killer cells, and B cells. Acute rejection is associated with a high incidence of complications, such as infection and graft vs host disease.
- A single episode of acute rejection that is promptly diagnosed and treated often prevents organ or tissue failure. Recurrence of acute rejection can lead to chronic rejection.

Chronic rejection: most often progresses gradually over several years

- Leads to fibrosis and loss of graft function
- Mechanism not well-understood

LWW Health Library

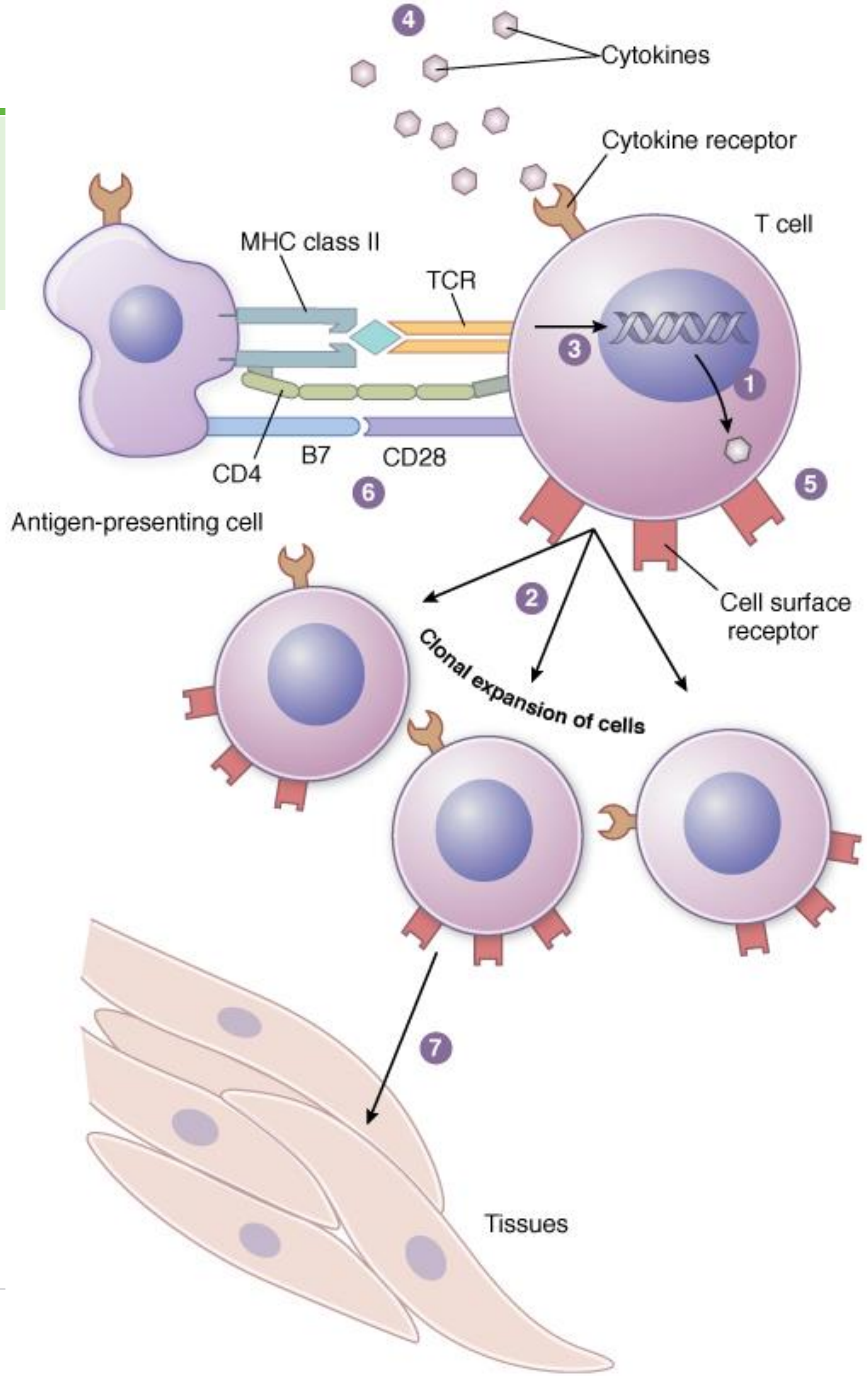
From: **46 Pharmacology of Immunosuppression**

Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy Fourth Edition, 4e, 2017; Fig 46-1

Pharmacologic Suppression of the Immune System

Overview of mechanisms of pharmacologic immunosuppression.

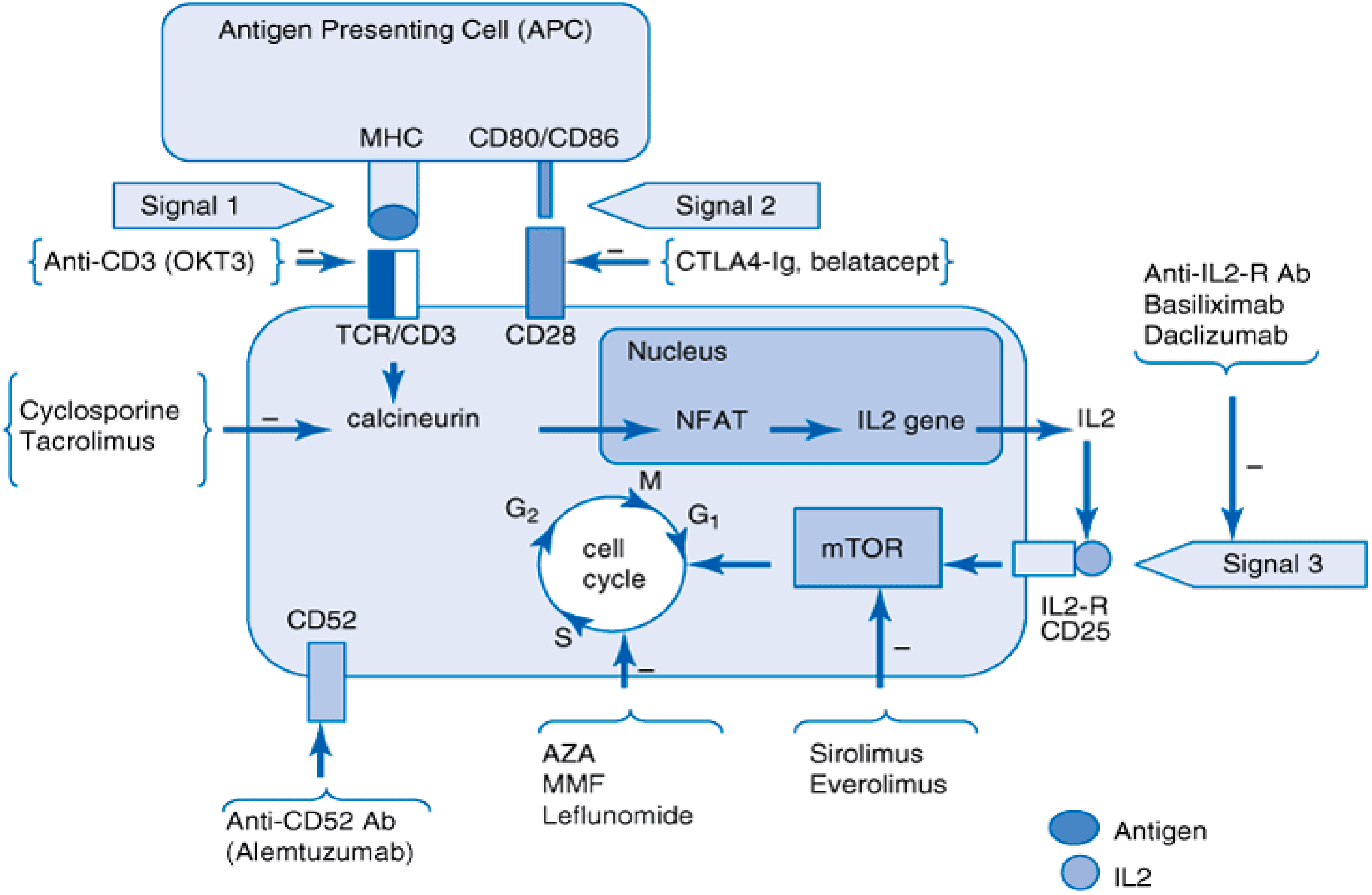
1) Inhibit gene expression of pro-inflammatory mediators
2) Attack on expanding lymphocyte population
3) Intracellular signaling inhibition
4) Cytokine and cytokine receptor neutralization
5) Selective depletion of immune cells
6) Block of costimulation
7) Block of lymphocyte-target interaction
8) Suppression of innate immune cells and complement activation (not shown)



Copyright © Wolters Kluwer

Date of download: 4/6/2022

T Lymphocyte-Related Targets of Transplant Immunosuppressive Agents



Source: Lerma EV, Berns JS, Nissenson AR: *CURRENT Diagnosis & Treatment: Nephrology & Hypertension*: http://www.accessmedicine.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Immunosuppressive therapies of organ transplantation:

Induction therapy

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Drug	Mechanism	Effect
✳Antithymocyte globulin rATG	Purified polyclonal antibodies may bind a variety of T cell surface antigens, leading to depletion of T cells that circulate between the blood and lymph.	Profound T cell depletion Preferred agent for patients with high and low risk of acute rejection
✳Basiliximab IL-2R α chain (CD25) inhibitor	Prevents T cell proliferation and cytokine production	Non-depleting agent An option for patients with low risk of acute rejection.
✳Alemtuzumab CD52 inhibitor	Direct antibody-dependent lysis of T and B cells.	Potent and prolonged panlymphocyte (T and B cells) depletion An infrequently used alternative.

Immunosuppressive therapies of organ transplantation:

Maintenance therapy

Drug	Mechanism
Calcineurin inhibitors (CNIs) ✳Tacrolimus ✳Cyclosporine	Immunophylin-CNI-calcineurin complex prevents activation of NFAT Decreases synthesis of IL-2 and other cytokines
Antimetabolites ✳Mycophenolate mofetil ✳Azathioprine	Inhibit de novo purine (nucleic acid) synthesis, which inhibits DNA synthesis in T and B cells
Glucocorticoids ✳Methylprednisolone ✳Prednisone	Regulate gene expression. GC therapy rapidly reduces T cell populations by lysis or redistribution, which has suppressive effects on inflammatory cytokines and chemokines.
T cell costimulatory blocker ✳Belatacept (CTLA-4 fusion protein)	Belatacept CTLA-4 binds CD80/86 on antigen presenting cells, which prevents the costimulatory T cell activating signal: the binding of CD28 on T cells to CD80/86.
Proliferation signal inhibitors ✳Sirolimus Everolimus	Blocking the protein kinase mTOR interrupts progression from the G1 to S phase, causing cell cycle arrest, thus inhibiting cytokine mediated T cell proliferation.

Immunosuppressive therapies organ transplantation:
Established acute allograft rejection

T cell-mediated and/or antibody-mediated rejection based on histological evidence (biopsy): Requires the use of agents directed against activated T cells and/or B cells

GENERAL APPROACH

Acute T cell-mediated rejection

Goal: To reverse rejection and return allograft function

Grading system determines regimen and dosing.

- Methylprednisolone, IV high-dose pulse, then oral taper (monotherapy)
- Methylprednisolone and rATG
  - Alemtuzumab if patient does not tolerate rATG

Antibody-mediated rejection

Goal: To eradicate clonal population of plasma cells or B cells causing the rejection

Regimen determined by time of onset, <1 year, >1 year

- Methylprednisolone + intravenous immune globulin (IVIG) ± plasmapheresis
  - methylprednisolone, IV high-dose pulse, then oral taper
- Individual cases:
  - plasmapheresis / rituximab

Complications of Immunosuppressive Therapy

Infections: Bacterial, fungal, viral, parasitic

Risk factors: The intensity and duration of immunosuppression

- Vaccines should be administered prior to transplantation.

Malignancy: Post-transplant lymphoproliferative disorders (PTLD) is most common.

PTLD:

Post-transplant lymphoproliferative disorders are **B lymphocyte proliferations** that occur in patients receiving chronic immunosuppression (due to decreased T cell immune surveillance) for solid organ or allogeneic hematopoietic cell transplantation.

PTLD is an Epstein-Barr virus (EBV)-positive B cell proliferation in most patients.

Approximately 90 to 95 percent of adults show serologic evidence of infection.

Principle risk factors: The degree of T cell immune suppression and the EBV serologic status of the patient.

T lymphocyte depleting agent
Antithymocyte globulin
✳rATG [Thymoglobulin]

Rabbit-derived antithymocyte globulin (rATG) is comprised of purified IgG antibodies isolated from the serum of rabbits immunized with human thymocytes.

rATG is most frequently used depleting agent.

Equine antithymocyte globulin, ATGAM, is used less frequently.

Video: Mechanism of action by the manufacturer of Thymoglobulin
https://www.thymoglobulin.com/about?_gl=1*1jo03ka*_up*MQ..&gclid=d3a8c96dd93812774a3aac94782844ac&gclsrc=3p.ds#seeHowItWorks

rATG binds T cell antigens and produces profound lymphopenia.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic

- It appears that ATG acts on a wide variety T cell surface antigens, leading to depletion of T cells that circulate between the blood and lymph.
- CD2, CD3, CD4, CD8, CD11a, CD18, CD25, CD44, CD45, and HLA class I and II molecules
- Plasma t½ 2-3 days
- Lymphopenia may persist for up to 1 year.

Action: Depletion of peripheral T cells

- **Direct cytotoxicity:** complement-mediated and antibody-dependent cellular cytotoxicity
- **Block lymphocyte function:** bind surface molecules involved in the regulation of cell function
- **Results in marked suppression of T cell response to the allograft.**

rATG Adverse Effects

Infusion reactions, common:

- Flu-like symptoms:

Fever, chills, nausea, muscle/joint pain

- Infusion site pain

Rapid infusion rates have been associated with **cytokine release syndrome** and serious cardiopulmonary events.

Practice point:

Pretreatment with a glucocorticoid, acetaminophen, and/or an antihistamine may reduce infusion related reactions.

Class effects of immunosuppressants:

Infection, PTLD

Risk is related to intensity of therapy.

Pregnancy recommendation:

People of reproductive age should use effective contraception during and at least 3 months following treatment.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

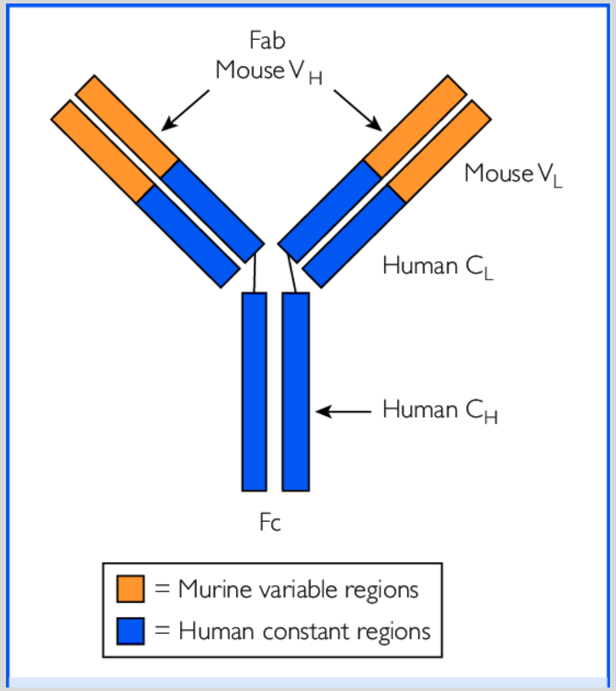
T cell nondepleting agent

✳Basiliximab

IL-2 receptor α chain inhibitor

An anti-CD-25 chimeric monoclonal antibody.

IL-2Rα = CD25



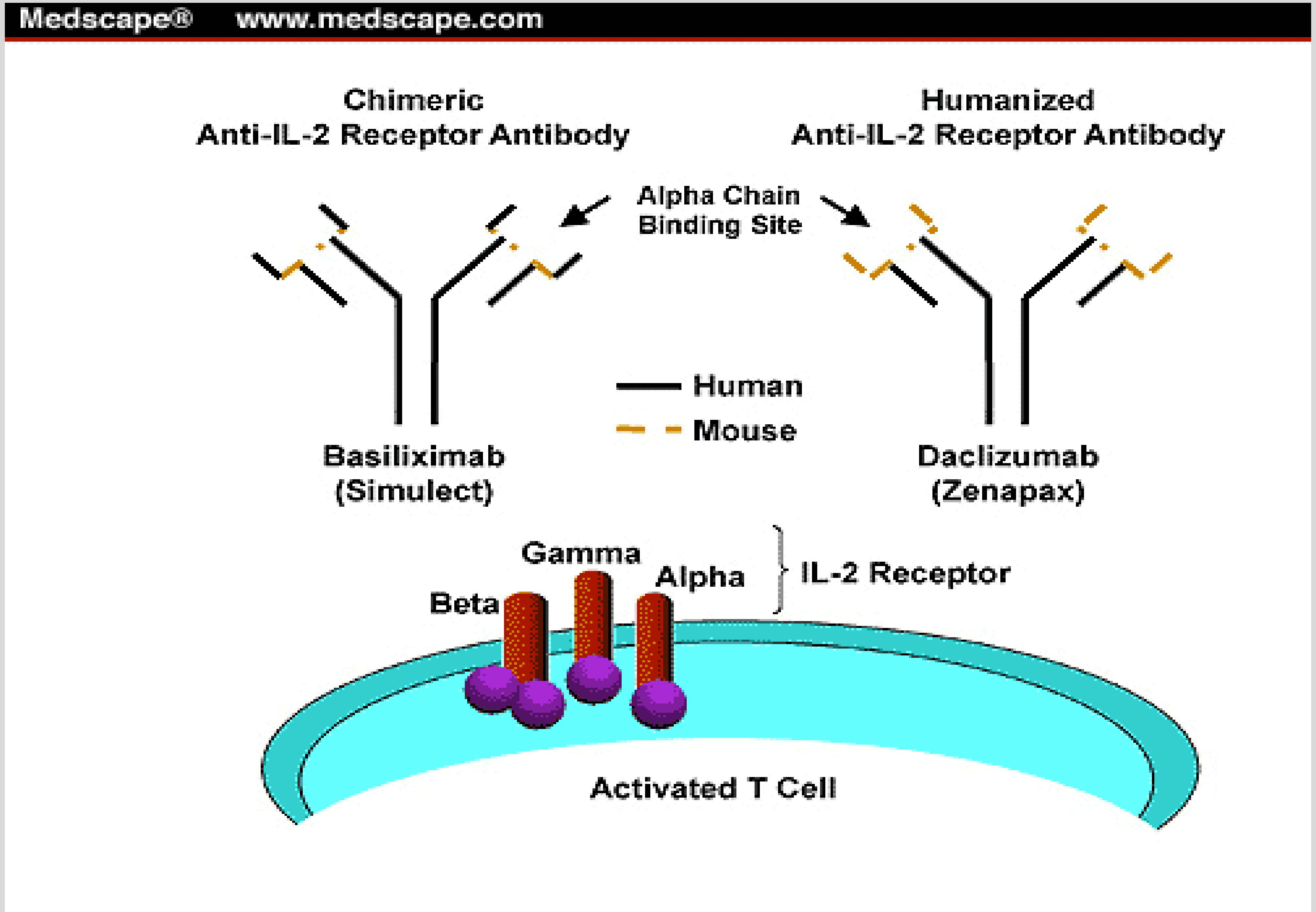
Chimeric antibody -xi-mab
Mouse variable domains
Human constant domains

Baxiliximab Mechanism of Action

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

1. high affinity binding to IL-2Rα on surface of activated T cells
2. competitively blocks IL-2 binding to activated T cells
3. prevents T cell activation
4. IL-2R downregulation prevents proliferation

Basiliximab does not cause lysis or destruction of T cells



http://img.medscape.com/fullsize/migrated/437/182/437182.fig3.gif

Figure 3. The IL-2 receptor antagonists.

Basiliximab Therapeutic Uses and Toxicities

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Uses

- Induction therapy in transplant patients with low risk of acute rejection

Together with IV methylprednisolone and maintenance immunosuppression

kidney, heart, liver, lung

2 doses (day 0, day 4)

- Tx acute refractory GVHD

Generally well tolerated

Potential Toxicities

- Hypersensitivity reactions

Note:

- No cytokine release syndrome
- No significant metabolic drug interactions reported

Panlymphocyte depleting agent

✳Alemtuzumab

Anti-CD52 monoclonal antibody

Alemtuzumab (Campath/Lemtrada) is available only through special programs.

Alemtuzumab’s apparent mechanism of action

binds CD52 on T and B cells

↓

induces direct lysis of cells

↓

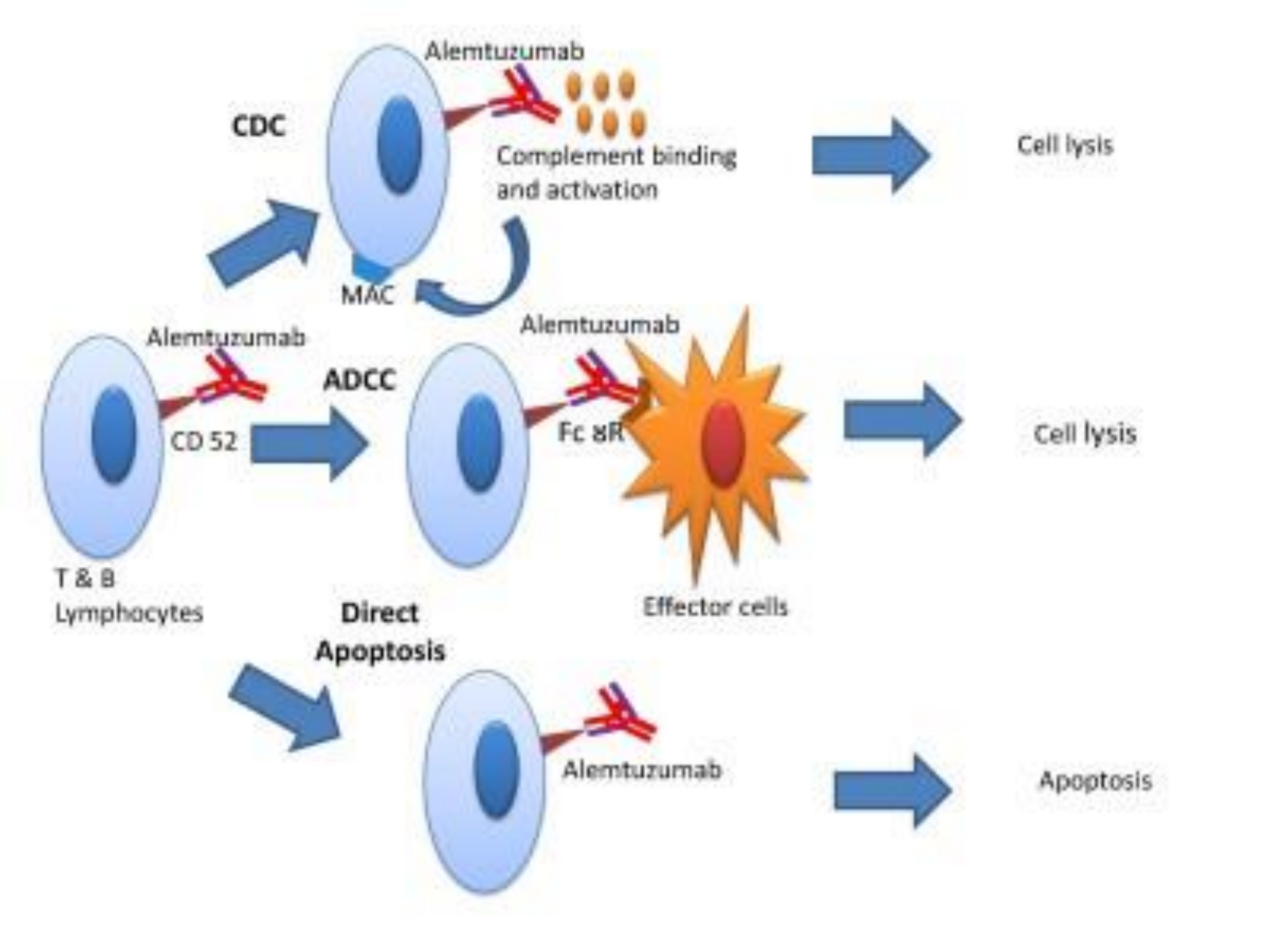
depletion of the T and B cells

↓

reduces host immune response against the transplanted organ

↓

profound, prolonged immunosuppression (serious and fatal infectious complications)



Dove Press (open access) Intractable and highly active relapsing multiple sclerosis – role of alemtuzumab
https://www.dovepress.com/intractable-and-highly-active-relapsing-multiple-sclerosis-ndash-role--peer-reviewed-fulltext-article-NDT

Alemtuzumab Adverse Effects (not a complete list)

associated with serious and life-threatening adverse effects

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Hematologic effects

lymphopenia, neutropenia, thrombocytopenia, autoimmune hemolytic anemia and autoimmune idiopathic thrombocytopenia, pancytopenia

Serious infections / malignancy:

Bacterial, viral, fungal, protozoal

Malignant neoplasms

Infusion reactions

Serious and life-threating reactions including anaphylaxis

Multiple sclerosis treatment:

Autoimmune effects

Malignancies

Serious infusion reactions

Pregnancy: For patients who could become pregnant, effective contraception is recommended during and at least 3 months after the last dose.

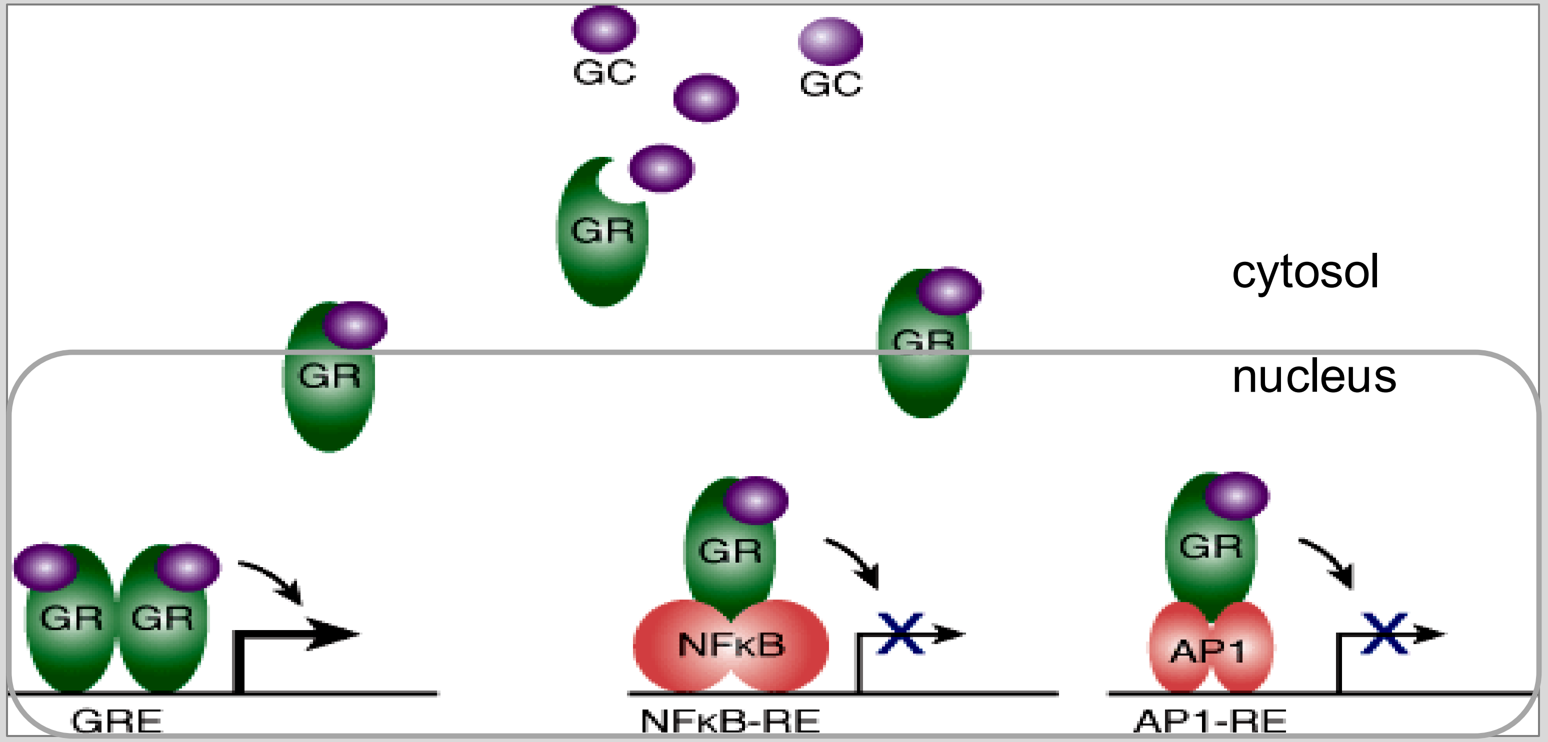
Glucocorticoids

- ✳Prednisone
- ✳Methylprednisolone

Numerous others

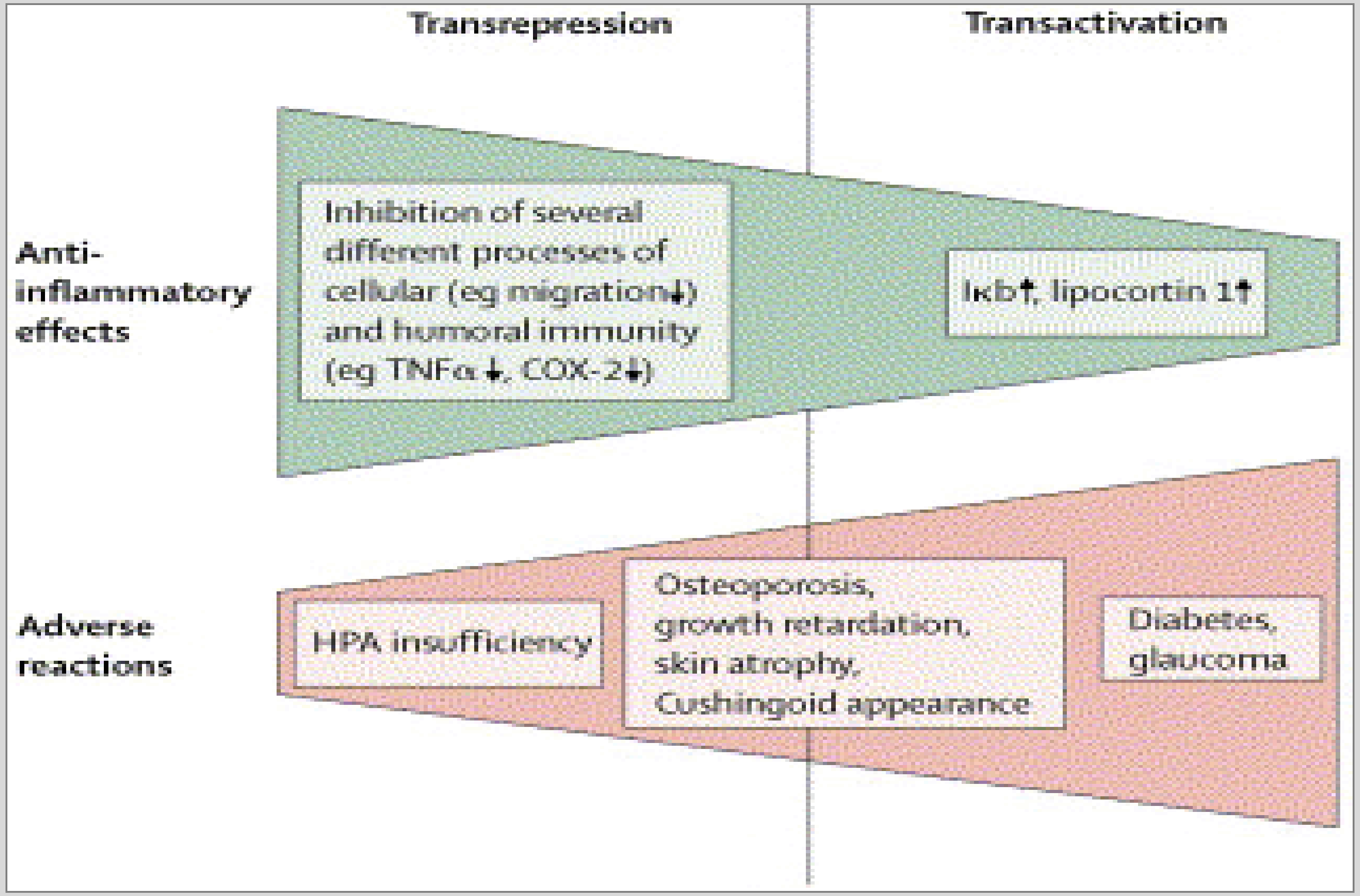
Binding of the GC-GR complex to nuclear elements → transrepression or transactivation of genes

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



transcription

transrepression



Left: Pathophysiology of Disease: An Introduction to Clinical Medicine, 7e; Figure 21-8

Glucocorticoids have multiple actions → broad anti-inflammatory effects

- Induce lipocortin → inhibits PLA2 → inhibits synthesis of prostaglandins and lipoxygenase products
- Inhibit T cell proliferation
- Humoral immunity dampened

- ↑ Lymphocyte apoptosis
- ↑ Lymphocyte redistribution
- ↓ IL-1; IL-6; IL-2
- ↓ T cell proliferation
- ↓ Chemotaxis
- ↓ Lysosomal enzyme release of neutrophils and monocytes
- ↓ synthesis of prostaglandins and lipoxygenase products

Glucocorticoids Therapeutic Uses

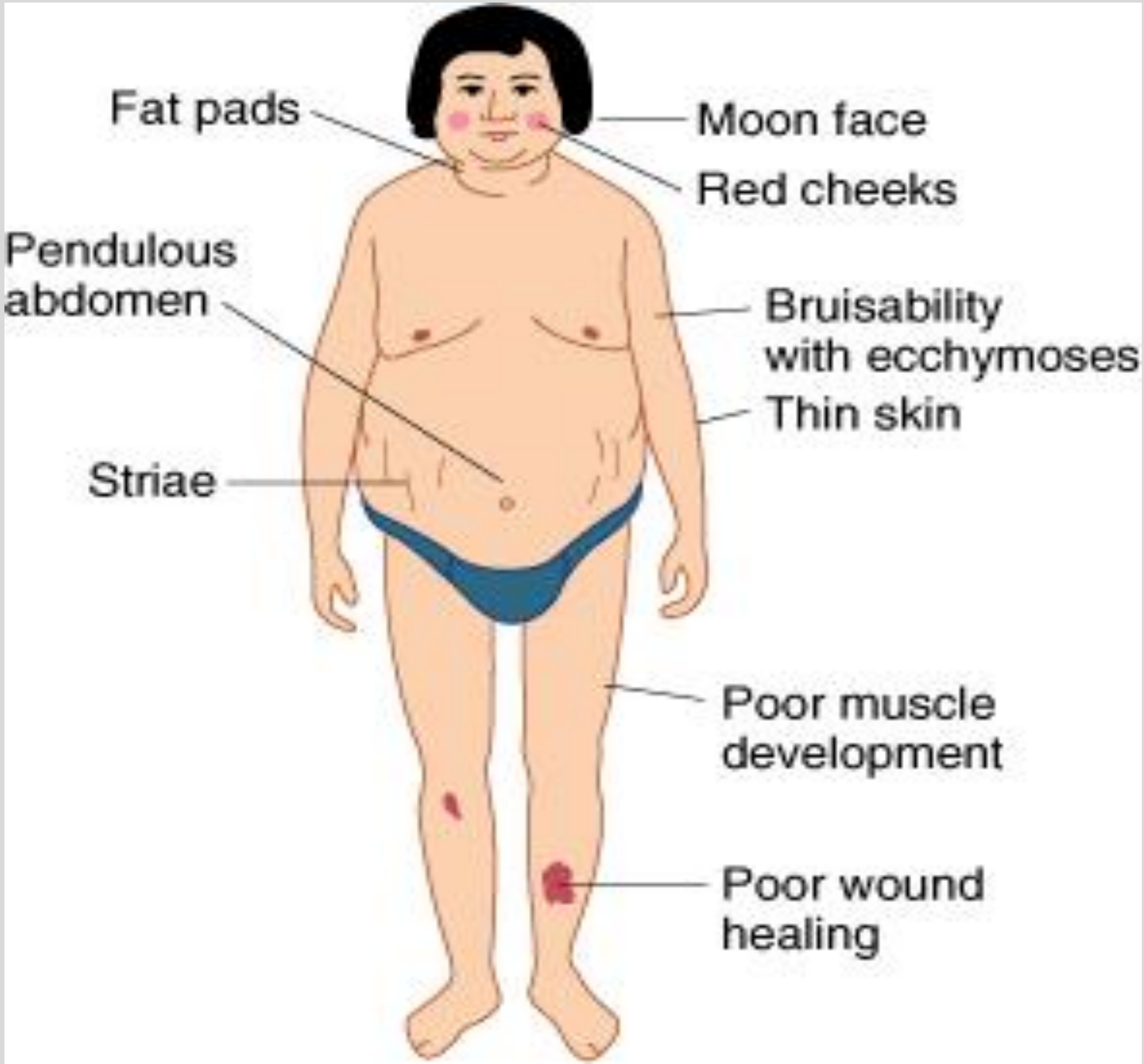
- Prevention and treatment of transplant rejection
  - Induction therapy: IV methylprednisolone with ATG or basiliximab
  - Maintenance prophylaxis of acute allograft rejection: oral prednisone
  - Acute transplant rejection: High-dose methylprednisolone pulse (intermittent) therapy
- GVHD prevention and treatment
- Management of cytokine release syndrome, infusion reactions

Glucocorticoids are used in a wide variety of medical conditions to suppress underlying undesirable immunologic reaction.

Dose-Related Adverse Effects of Glucocorticoids

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Acute Effects	Delayed Effects
Impaired immunity ↑ risk of opportunistic infections Delayed wound healing Hyperglycemia; diabetes ↑ Plasma cholesterol ↑ salt retention and ↑ BP GCs with mineralocorticoid action Mood – euphoria/depression	Peptic ulcers; GI bleeding Cushingoid effects: –Moon face; ruddy complexion –Nuchal fat deposition (buffalo hump) –Weight gain; abdominal obesity –Muscle wasting/weakness of limbs –Thin skin; bruisability; hirsuitism; acne Osteoporosis; Avascular necrosis of bone Cataracts Reduced growth rate in children
Higher doses are more likely to cause acute effects.	Greater cumulative doses are more likely to cause chronic adverse effects.



Cushingoid Effects

Glucocorticoids have numerous effects and act on most body tissues.

These actions on body tissues are demonstrated by the clinically observed effects.

Review of Medical Physiology, Ganong's - 23rd Ed. (2010) Figure 22-13. Typical findings in Cushing syndrome. (Reproduced with permission from Forsham PH, Di Raimondo VC: *Traumatic Medicine and Surgery for the Attorney*. Butterworth, 1960.)

Summary of induction agents and glucocorticoids in transplantation immunosuppression

Link to a mind map (concept map) of immunosuppressants in transplant pharmacology:

https://www.researchgate.net/figure/Mechanisms-of-action-and-targets-of-immunosuppressive-drugs-used-in-renal_fig1_273503794

General approach to organ transplantation therapy

Patient preparation	Selection of the best available ABO blood type-compatible HLA match for organ donation
Combination immunosuppressant regimens	Drugs with different molecular targets → lower doses → maximize effects + limit toxicities
Intensive induction	The risk of rejection is highest immediately after the transplant. Thus, typically, a very potent anti-lymphocyte agent is used initially along with triple therapy for maintenance of acute rejection prophylaxis.
Maintenance therapy	Gradually taper down each agent in the initial triple therapy over several weeks to months to a maintenance level for prevention of rejection.
Follow-up	Careful evaluation of organ function
Rejection	Diagnosed by biopsy Rejection may be treated with high-dose “pulse” glucocorticoid therapy over several days or with antilymphocyte antibodies. Rejection therapy is effective in more than 90% of cases

- The principal approach to immunosuppressive therapy is to alter lymphocyte function using drugs or therapeutic antibodies against immune proteins.
- Immunosuppressive agents have been developed that selectively inhibit rejection of transplanted tissues.
- Immunosuppressants are also used to inhibit aggressive, aberrant immune response, and, thus, reduce the morbidity and mortality that result from autoimmune diseases and chronic inflammatory disorders.

In transplantation, the major classes of immunosuppressive drugs used today are:

- biologicals (antibodies)
- glucocorticoids
- calcineurin inhibitors
- antimetabolites
- costimulation blockers
- proliferation signal inhibitors

- Immunosuppressants share the common side effect of increasing susceptibility to infectious diseases.
- Long-term immunosuppressive therapies are associated with the development of malignancy.
- Post-transplant lymphoproliferative disorder (PTLD) is the most common malignancy complicating solid organ transplantation. The principal risk factors are the degree of T cell immunosuppression and Epstein-Barr serologic status of the patient.
- Antithymocyte globulins (ATG) are prepared by immunization of rabbits with human thymocytes, producing polyclonal IgG antibodies directed against T cells. rATG results in T cell depletion with profound lymphopenia.
- rATG, along with other agents, is used at the time of transplantation to prevent early allograft rejection, for the treatment of severe glucocorticoid-resistant acute rejection.

- Basiliximab is a chimeric murine/human monoclonal antibody that binds IL-2 receptor α chain (CD25) on activated T cells, which interferes with T cell proliferation. It is a non-depleting agent. It is used for induction in patients with a low risk of early acute rejection in combination with a IV methylprednisolone. Maintenance immunosuppressive therapy is initiated at the same time. Basiliximab has a serum half-life of about 7 days and is generally well tolerated.
- Alemtuzumab is a humanized monoclonal antibody that binds CD52 in T and B cells, resulting in depletion of both cell lines. It produces profound, prolonged immunosuppression and is associated with serious and fatal infusion reactions and infections. It is infrequently used as alternative induction agent. It is approved for use in multiple sclerosis and B cell chronic lymphocytic leukemia. Alemtuzumab is not readily available.
- Glucocorticoids are mainstays for suppressing acute rejection in solid organ allografts and chronic GVHD. GCs are associated with numerous dose-related adverse effects.

References

- Access Medicine Goodman and Gilman’s The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 39: Immunosuppressants, Tolerogens, and Immunostimulants
- Access Medicine Katzung’s Basic & Clinical Pharmacology, 16e, 2024; Chapter 55: Immunopharmacology
- Lexicomp monographs
- UpToDate various articles, literature reviews are current.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R